EXPERIMENT - 21

21. Implement the Opening technique as a Morphological operation to dilate the foreground regions based on Open CV.

Aim

To implement the Opening technique as a morphological operation using OpenCV in Python to remove small noise from the foreground regions of an image.

Procedure

1. Import the OpenCV and NumPy libraries.
2. Read the input image using cv2.imread().
3. Resize the image to fit the screen.
4. Convert the image into grayscale.
5. Create a kernel using np.ones().
6. Apply the Opening operation using cv2.morphologyEx() with cv2.MORPH_OPEN.
7. Display the original image and the opening output image.

Program

```
import cv2
import numpy as np

img = cv2.imread("image.jpg")

if img is None:
    print("Image not found")
else:
    img = cv2.resize(img, (500, 350))
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    kernel = np.ones((5, 5), np.uint8)

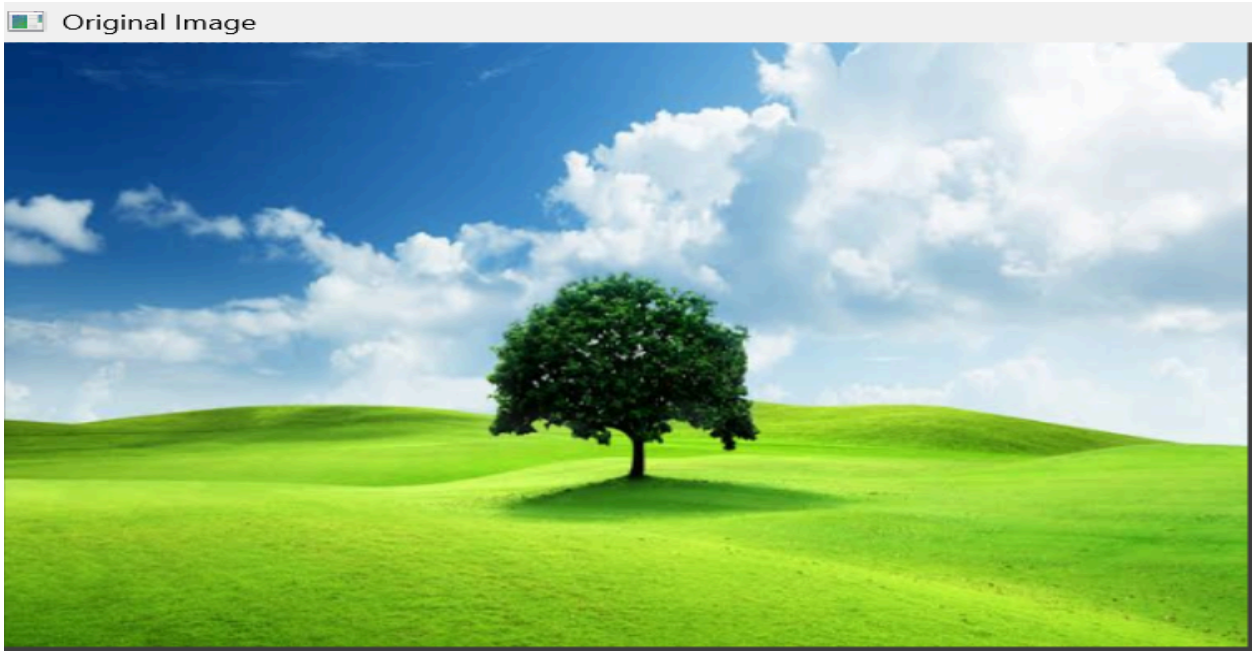
    opening = cv2.morphologyEx(gray, cv2.MORPH_OPEN, kernel)

    cv2.imshow("Original Image", img)
    cv2.imshow("Opening Output", opening)

    cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input Image



Output Image



Result

The Opening Morphological Operation was successfully implemented using OpenCV in Python to remove small noise from the image.